

Product Customization Canvas Design

Full Project Documentation, Architecture & Database Design

Complete & Unabridged Version

Part 1: Product Customization Canvas Design

১. Top Header

Header-এর মধ্যে থাকবে:

- Store logo
- Product name
- Selected variant
- Undo button
- Redo button
- Reset design
- Save design
- Preview button
- Add to Cart button
- Dynamic product price

২. Left-Side Toolbar

Upload Image

- JPG, PNG, SVG, WEBP upload
- Drag-and-drop upload
- Transparent PNG support
- Image size validation
- Low-resolution warning
- File size limit
- Uploaded image library
- Remove background option

Add Text

- Heading text
- Paragraph text
- Curved text
- Multiple text lines

- Custom fonts
- Google fonts
- Font size
- Bold, italic, underline
- Letter spacing
- Line height
- Text alignment
- Text color
- Text outline
- Text shadow

Graphics and Clipart

- Icons
- Logos
- Stickers
- Shapes
- Decorative elements
- Category-based clipart
- Search option
- Admin-uploaded graphics

Shapes

- Circle
- Rectangle
- Square
- Triangle
- Line
- Custom SVG shape
- Shape fill color
- Border color
- Border width
- Corner radius

Other Tools

- QR code generator
- Barcode generator

- Pattern selector
- Monogram generator
- Background color
- Product color selector

৩. Main Product Canvas

Canvas-এর মাঝখানে product mockup থাকবে।

প্রয়োজনীয় Canvas feature:

- Drag element
- Resize element
- Rotate element
- Delete element
- Duplicate element
- Lock element
- Flip horizontally
- Flip vertically
- Bring forward
- Send backward
- Center horizontally
- Center vertically
- Multi-element selection
- Keyboard arrow movement
- Copy and paste
- Snap-to-center guideline
- Smart alignment guideline
- Zoom in and zoom out
- Fullscreen canvas
- Mouse wheel zoom
- Touchscreen gesture support

Canvas তৈরি করার জন্য Fabric.js ব্যবহার করা সবচেয়ে ভালো হবে।

৪. Printable Area

Product-এর উপর একটি নির্দিষ্ট printable area থাকবে।

যেমন:

T-shirt chest print area

Mug front print area

Bottle label area

Phone case design area

Printable area feature:

- Safe-area boundary
- Bleed-area boundary
- Out-of-area warning
- Print-area clipping
- Element canvas-এর বাইরে গেলে warning
- Printable dimension display
- CM, MM, Inch অথবা Pixel measurement
- Admin থেকে print area configure করার সুবিধা

৫. Right-Side Settings Panel

User যে element select করবে, সেই অনুযায়ী settings দেখাবে।

Text Selected হলে

- Font family, Font size, Text color, Font weight, Alignment, Letter spacing, Line height, Outline, Shadow, Opacity, Rotation

Image Selected হলে

- Width, Height, Rotation, Opacity, Crop, Mask, Filters, Brightness, Contrast, Saturation, Remove background, Replace image

Shape Selected হলে

- Fill color, Border color, Border width, Opacity, Corner radius, Rotation

৬. Layer Management

একটি Layers panel থাকা দরকার।

- সব element-এর list
- Layer reorder
- Layer hide/show
- Layer lock/unlock

- Layer rename
- Duplicate layer
- Delete layer
- Bring to front
- Send to back

৭. Multiple Product Views

Product অনুযায়ী একাধিক customization view থাকবে।

উদাহরণ:

- **T-shirt:** Front, Back, Left sleeve, Right sleeve
- **Mug:** Front, Back, Full wrap
- **Bottle:** Front, Back, Full label wrap

প্রতিটি view-এর design আলাদাভাবে save হবে।

৮. Product Options

Canvas-এর নিচে বা side panel-এ থাকবে:

- Product color
- Size
- Material
- Style
- Print location
- Quantity
- Product mockup type
- Variant selection

Variant change করলে canvas design যেন হারিয়ে না যায়।

৯. Preview System

User final product preview দেখতে পারবে।

- Front preview
- Back preview
- Different angle preview
- Fullscreen preview

- Product color preview
- Realistic mockup preview
- Downloadable preview
- Mobile preview
- Before/after design comparison

Future version-এ 3D preview যুক্ত করা যাবে।

১০. Dynamic Pricing

Customization অনুযায়ী price update হবে।

উদাহরণ:

- Base product price
- Extra print area charge
- Premium font charge
- Background removal charge
- Multiple-side print charge
- Large print charge
- Quantity discount
- Custom graphic charge

Price calculation:

```
Base Product Price
+ Customization Charge
+ Additional Print Side
+ Special Material
- Quantity Discount
= Final Price
```

১১. Save Design

User design save করতে পারবে।

- Autosave
- Save design manually
- Guest design save
- Logged-in user design library
- Generate unique design ID

- Copy design link
- Edit saved design
- Reorder saved design
- Design expiration control

১২. Shopify Add to Cart Integration

Add to Cart করার সময় Shopify-তে পাঠাতে হবে:

- Product ID
- Variant ID
- Quantity
- Product size
- Product color
- Design ID
- Preview image URL
- Print-ready file URL
- Selected customization options
- Additional price
- Custom text information

Cart item properties-এ এই information দেখানো যাবে।

১৩. Export System

Admin-এর জন্য export feature অত্যন্ত গুরুত্বপূর্ণ।

- High-resolution PNG
- Transparent PNG
- SVG
- PDF
- Print-ready file
- 300 DPI output
- Separate export for each product side
- Design JSON file
- Preview mockup
- Customer artwork download
- Order-wise export

১৪. Mobile Responsive Design

Mobile features:

- Bottom toolbar
- Swipe between product views
- Pinch-to-zoom
- Touch drag
- Touch resize
- Collapsible settings drawer
- Sticky Add to Cart
- Full-width canvas
- Mobile-safe controls

১৫. Admin Dashboard Features

Admin dashboard থেকে নিয়ন্ত্রণ করা যাবে:

- Products connect করা
- Shopify products sync করা
- Mockup upload করা
- Print area define করা
- Front/back view configure করা
- Product color set করা
- Clipart upload করা
- Font manage করা
- Customization price set করা
- Design template create করা
- User designs দেখা
- Order design download করা
- Print-ready file download করা
- File size এবং resolution rules set করা

১৬. প্রয়োজনীয় Advanced Features

- Design template
- Pre-made layouts
- AI background remover

- AI image generator
- Image quality checker
- Spell checker
- Text overflow warning
- Low-resolution warning
- Auto-center design
- Auto-fit design
- Design version history
- Real-time autosave
- Customer design sharing
- Duplicate previous order
- Bulk quantity pricing
- Multilingual interface
- Currency support
- Accessibility support

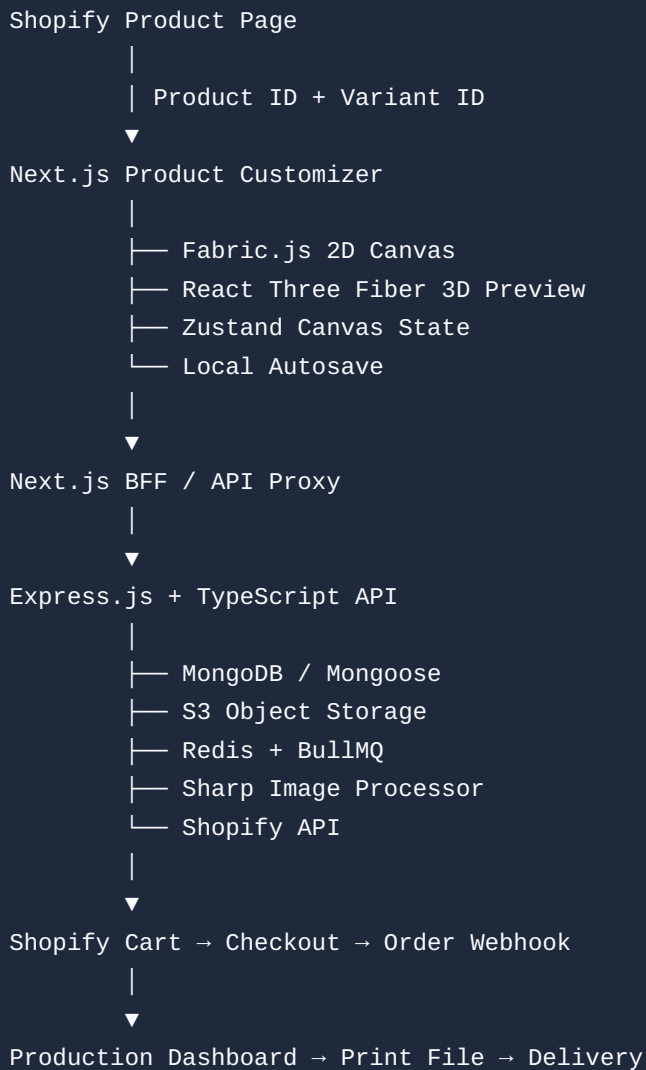
Recommended Canvas Structure

সবচেয়ে clean এবং professional structure হবে:

- Desktop: Left tools + Middle canvas + Right settings
- Mobile: Full-width canvas + Bottom tools + Slide-up settings
- Bottom area: Product views, variants এবং pricing
- Top area: Undo, redo, preview, save এবং Add to Cart

এই structure দিয়ে T-shirt, mug, bottle, pouch, phone case, signage এবং অন্যান্য customizable product একই system-এর মধ্যে manage করা যাবে।

Part 2: Recommended Advanced MERN Architecture



5. Frontend Technology

Next.js + React + TypeScript

ব্যবহার করবেন:

- Next.js App Router
- React
- TypeScript
- Tailwind CSS
- shadcn/ui
- Zustand বা Redux Toolkit
- React Query/TanStack Query

- React Hook Form
- Zod validation

Responsibility: Product customizer interface, Product ও variant selection, Canvas toolbar, Settings panel, Layers panel, Save design, Preview, Add to cart, Customer design library, Admin dashboard.

Component structure

```
apps/web/
├── app/
│   ├── customize/[productHandle]/
│   ├── designs/
│   ├── admin/
│   └── api/
├── components/
│   ├── customizer/
│   │   ├── CanvasEditor.tsx
│   │   ├── ToolsPanel.tsx
│   │   ├── PropertiesPanel.tsx
│   │   ├── LayersPanel.tsx
│   │   ├── ProductViews.tsx
│   │   └── PriceSummary.tsx
│   └── ui/
├── stores/
├── hooks/
└── services/
```

২. Canvas Technology

Fabric.js ব্যবহার করবেন। Product customization-এর 2D editor-এর জন্য Fabric.js সবচেয়ে উপযোগী হবে। এটি support করে:

- Object selection, Drag and drop, Resize, Rotate, Layer ordering
- Text editing, Shapes, SVG, Clip paths, Image filters
- Canvas state serialization, JSON save/load, SVG এবং image export

Fabric.js interactive canvas, object transformation, stack ordering, serialization, JSON/SVG/image export এবং viewport handling দেয়। (Fabric.js)

Canvas Engine আলাদা package করুন

```
packages/canvas-engine/  
├─ core/  
│   ├─ createCanvas.ts  
│   ├─ loadDesign.ts  
│   ├─ saveDesign.ts  
│   └─ exportDesign.ts  
├─ objects/  
│   ├─ addText.ts  
│   ├─ addImage.ts  
│   ├─ addShape.ts  
│   └─ addClipart.ts  
├─ controls/  
├─ history/  
├─ guidelines/  
├─ print-area/  
└─ types/
```

এতে frontend এবং export worker একই canvas logic ব্যবহার করতে পারবে।

৩. Optional 3D Preview

Realistic T-shirt, mug, bottle বা pouch preview-এর জন্য:

- Three.js
- React Three Fiber
- Drei
- GLTF/GLB product models
- Dynamic CanvasTexture

React Three Fiber হলো Three.js-এর React renderer। এটি দিয়ে React component-এর মতো 3D scene, mesh, material এবং texture manage করা যায়। (Poimandres)

3D flow

```
Fabric.js Canvas
|
| Export preview texture
▼
CanvasTexture
|
▼
Three.js Product Material
|
▼
Real-time 3D Product Preview
```

প্রথম version-এ 2D mockup রাখবেন। 3D preview Phase 3-এ যোগ করাই ভালো।

8. Backend Technology

Express.js + TypeScript

Express API business logic handle করবে:

- Authentication, Product sync, Design save, Asset metadata, Pricing calculation, Export jobs, Shopify cart creation, Shopify webhooks, Order processing, Admin permissions

Express routing ও middleware-based framework হওয়ায় authentication, validation, rate limiting এবং role permissions আলাদা middleware হিসেবে রাখা যায়। Express 5 async handler-এর rejected Promise error middleware-এ পাঠাতে পারে। (Express.js)

Backend structure

```
apps/api/
├── src/
│   ├── modules/
│   │   ├── auth/
│   │   ├── products/
│   │   ├── templates/
│   │   ├── designs/
│   │   ├── assets/
│   │   ├── pricing/
│   │   ├── exports/
│   │   ├── carts/
│   │   ├── orders/
│   │   └── webhooks/
│   ├── middleware/
│   ├── services/
│   ├── queues/
│   ├── sockets/
│   ├── config/
│   └── server.ts
```

৫. MongoDB Database Structure

Recommended collections: users, products, customizerTemplates, designs, designVersions, assets, exportJobs, customOrders, webhookEvents.

MongoDB transaction ব্যবহার করে design, version এবং asset metadata-এর মতো related update atomicভাবে করা যাবে। Change Streams দিয়ে design export বা order status পরিবর্তন হলে real-time event তৈরি করা সম্ভব। (MongoDB)

৬. Step-by-Step Complete Work Flow

Step 1: Shopify Product Sync

Admin app install বা connect করার পরে Express backend Shopify থেকে সংগ্রহ করবে: Product, Variants, Images, Price, Options, Inventory, Metafields.

MongoDB-এর products collection-এ Shopify ID সংরক্ষণ করবেন।

Sync trigger: App installation, Manual sync, Product update webhook, Product delete webhook

Step 2: Admin Customizer Setup

Product

- └─ Front mockup
- └─ Back mockup
- └─ Left-side mockup
- └─ Right-side mockup
- └─ Print area
- └─ Safe area
- └─ Bleed area
- └─ Allowed tools
- └─ Fonts
- └─ Cliparts
- └─ Pricing rules

Print area save করবেন normalized coordinate হিসেবে:

```
{
  "x": 0.22,
  "y": 0.18,
  "width": 0.56,
  "height": 0.64
}
```

এভাবে desktop, mobile এবং high-resolution export-এ একই print-area ratio রাখা যাবে।

Step 3: Customer Opens Customizer

Shopify product page-এর Customize button user-কে পাঠাবে:

```
https://customizer.example.com/customize/t-shirt
?product=gid://shopify/Product/123
&variant=gid://shopify/ProductVariant/456
```

Frontend backend থেকে load করবে: Product information, Selected variant, Mockups, Print areas, Allowed customization tools, Existing design, Pricing rules.

Step 4: Initialize Canvas

- Fabric Canvas তৈরি করবে।
- Product mockup background হিসেবে বসাবে।
- Print area clip path তৈরি করবে।

- Safe-area guideline দেখাবে।
- Product view অনুযায়ী canvas state load করবে (Front View → frontCanvasJSON ইত্যাদি)।

Step 5: Customer Customizes Product

Canvas tools (Upload, add text, etc.)। প্রতিটি object-এর custom metadata রাখবেন:

```
{
  "id": "object_uuid",
  "type": "image",
  "assetId": "asset_id",
  "view": "front",
  "printable": true,
  "locked": false
}
```

Step 6: Canvas State Management

Zustand store-এর মধ্যে পুরো Fabric object রাখবেন না। Store-এ রাখবেন: Selected object ID, Active product view, Zoom level, Undo/redo state, Dirty state, Pricing selections, Canvas JSON snapshot।

Step 7: Image Upload Pipeline

```
Customer selects file
    ↓
Frontend validates size/type
    ↓
Express generates presigned upload URL
    ↓
Browser uploads directly to S3
    ↓
Express saves asset metadata
    ↓
Processing job starts
    ↓
Optimized image returned
```

Upload validations: MIME type, File extension, Magic bytes, Max size, Dimensions, Transparent PNG, SVG sanitization, Virus scanning, hash detection.

Step 8: Client-Side Image Processing

ছোট operations client-side করা যায়: Thumbnail generation, Crop preview, Brightness preview, Contrast preview, Background preview, Low-resolution warning। Heavy processing Web Worker-এ পাঠাবেন।

Step 9: Autosave System

Local autosave (IndexedDB) এবং Server autosave (Send compressed JSON)।

Step 10: Preview Generation

Instant preview (browser PNG/WebP) এবং Production preview (Backend worker high-resolution rendering)।

Step 11: Print-Ready Export

Print width ও desired DPI থেকে output pixels calculate করবেন:

```
Pixel Width  = Print Width in Inches × DPI
Pixel Height = Print Height in Inches × DPI
```

```
Production manifest
{
  "designId": "DES-10001",
  "product": "Premium T-Shirt",
  "variant": "Black / XL",
  "printMethod": "DTG",
  "views": [
    {
      "name": "front",
      "widthInches": 12,
      "heightInches": 16,
      "dpi": 300,
      "file": "print-front.png"
    }
  ]
}
```

৭. Background Job Technology

Redis + BullMQ। এই কাজগুলো API request-এর মধ্যে করবেন না: High-resolution export, Image optimization, Background removal, Mockup generation, PDF generation, Shopify product bulk sync, Email notification।

```
Main Export Job
├─ Front Print Export
├─ Back Print Export
├─ Preview Generation
└─ Production Manifest
```

৮. Dynamic Pricing System

MongoDB-তে pricing rules রাখবেন। Price অবশ্যই backend থেকে calculate করবেন।

```
{
  "basePrice": 25,
  "rules": [
    { "type": "additional_view", "amount": 5 },
    { "type": "background_removal", "amount": 3 },
    { "type": "premium_material", "amount": 8 }
  ]
}
```

৯. Add to Cart Flow

```
Customer clicks Add to Cart
    ↓
Validate design
    ↓
Create locked design version
    ↓
Create export job
    ↓
Generate preview
    ↓
Calculate final price
    ↓
Add Shopify merchandise line
    ↓
Attach design information
    ↓
Redirect/open Shopify cart
```

১০. Shopify Order Webhook Flow

```
Shopify Order Created Webhook
↓
Verify Webhook HMAC
↓
Check Event Idempotency
↓
Read Design ID
↓
Create CustomOrder
↓
Lock Design
↓
Generate Final Print Files
↓
Notify Production Admin
```

১১. Production Dashboard

Admin দেখতে পারবে: New customization orders, Design preview, Front/back print files, Customer notes, Product variant, Print dimensions, Resolution warnings, Export job progress, Payment status, Production status, Shipping status।

Order status: New → Design Validation → File Preparation → Ready for Production → In Production → Quality Check → Packed → Shipped → Delivered

১২. Real-Time Status

Real-time update-এর জন্য ব্যবহার করতে পারেন: Socket.IO, Server-Sent Events, MongoDB Change Streams, BullMQ QueueEvents।

১৩. Security Features

HTTP-only secure cookie, JWT/session rotation, Role-based permissions, Zod request validation, Rate limiting, Helmet security headers, Strict CORS, Shopify webhook HMAC verification, File signature validation, SVG sanitization, Presigned upload URLs, Short-lived download URLs, Idempotency key, Audit logs, Admin 2FA, Private S3 bucket।

8. Performance Optimization

- **Frontend:** Canvas editor dynamic import, Disable SSR for canvas, Memoized toolbar, Debounced autosave, Lazy-load cliparts, Web Worker processing, Mobile reduced preview.
- **Backend:** Redis cache, BullMQ workers, MongoDB indexes, S3/CDN assets, Cursor pagination, Compressed Canvas JSON, Idempotency, API vs image workers.

9. Monorepo Structure

```
product-customizer/  
├─ apps/  
│   ├─ web/           # Next.js  
│   ├─ api/           # Express.js  
│   └─ worker/        # BullMQ + Sharp  
├─ packages/  
│   ├─ database/      # Mongoose models  
│   ├─ canvas-engine/ # Fabric.js logic  
│   ├─ shared-types/  # TypeScript types  
│   ├─ validation/    # Zod schemas  
│   ├─ shopify/       # Shopify API client  
│   └─ pricing-engine/  
├─ docker/  
├─ package.json  
└─ turbo.json
```

10. Development Phase Plan

- **Phase 1 — MVP:** Shopify product connect, Product/variant selection, Fabric.js canvas, Upload image, Add/edit text, Resize/rotate/delete, Front/back views, Print area, Canvas JSON save, Preview generation, Dynamic price, Shopify add to cart, Order webhook, Admin print-file download.
- **Phase 2 — Production Ready:** Autosave, Design versioning, Redis/BullMQ, S3 direct upload, High-resolution export, Resolution validation, Layers panel, Template system, Saved design library, Order production statuses, Real-time job progress, Error monitoring, Audit logs, Webhook idempotency.
- **Phase 3 — Advanced Technology:** Real-time 3D preview, AI background removal, AI image generation, Smart object alignment, Automatic design quality checking, Collaborative editing, QR-code generator, Curved text, Variable-data personalization, Multi-language support, Mobile gesture editor, Automated production routing.

Final Recommended Stack

Frontend: Next.js + TypeScript + Tailwind + shadcn/ui

Canvas: Fabric.js

3D: Three.js + React Three Fiber + Drei

State: Zustand + TanStack Query

Backend: Express.js + TypeScript

Database: MongoDB Atlas + Mongoose

Storage: Amazon S3 / Cloudflare R2

Queue: Redis + BullMQ

Image Processing: Sharp + SVG processing

Real-time: Socket.IO + MongoDB Change Streams

Commerce: Shopify Storefront API + Admin API + Webhooks

Deployment: Vercel frontend, Docker-based API and Worker, MongoDB Atlas, Managed Redis, S3-compatible storage

Part 3: Product Customizer Database Design

এই structure-টি Next.js + Express.js + MongoDB + Shopify + S3/R2 + Redis/BullMQ ভিত্তিক production-ready system-এর জন্য।

High-Level Database Architecture

Store

- └─ Store Members
- └─ Customers
- └─ Products
 - └─ Variants
 - └─ Customizer Config
 - └─ Product Views
 - └─ Print Areas
 - └─ Allowed Tools
 - └─ Pricing Rule Set
- └─ Design Resources
 - └─ Fonts
 - └─ Cliparts
 - └─ Templates
- └─ Designs
 - └─ Current View States
 - └─ Immutable Versions
 - └─ Version View States
- └─ Shopify Cart Sessions
- └─ Orders
 - └─ Order Items
 - └─ Locked Design Version
 - └─ Export Jobs
- └─ Assets
- └─ Webhook Events
- └─ Audit Logs

Recommended Collections (1 to 20)

1. stores

```
{
  _id: ObjectId,
  name: "Demo Fashion Store",
  shopDomain: "demo-store.myshopify.com",
  shopifyShopId: "gid://shopify/Shop/123456",
  accessTokenEncrypted: "encrypted-value",
  scopes: ["read_products", "write_products", "read_orders"],
  currency: "USD",
  timezone: "America/New_York",
  plan: { name: "professional", status: "active", trialEndsAt: null },
  settings: {
    autosaveInterval: 4000,
    defaultDpi: 300,
    maxUploadSizeMb: 25,
    allowGuestDesigns: true,
    designExpirationDays: 30
  },
  status: "active",
  installedAt: Date,
  uninstalledAt: null,
  createdAt: Date,
  updatedAt: Date
}
// Indexes: { shopDomain: 1 }, { shopifyShopId: 1 }, { status: 1 }
```

2. users

```
{
  _id: ObjectId,
  name: "John Doe",
  email: "john@example.com",
  passwordHash: "...",
  authProvider: "credentials",
  providerUserId: null,
  avatarUrl: null,
  emailVerifiedAt: Date,
  status: "active",
  lastLoginAt: Date,
  createdAt: Date,
  updatedAt: Date
}
```

3. store_members

```
{
  _id: ObjectId,
  storeId: ObjectId,
  userId: ObjectId,
  role: "admin",
  permissions: ["products.read", "products.write", "designs.read", "orders.process",
"settings.write"],
  status: "active",
  createdAt: Date,
  updatedAt: Date
}
```

4. customers

```
{
  _id: ObjectId,
  storeId: ObjectId,
  userId: ObjectId,
  shopifyCustomerId: "gid://shopify/Customer/123",
  guestSessionId: null,
  firstName: "David",
  lastName: "Smith",
  email: "david@example.com",
  phone: null,
  totalDesigns: 8,
  totalOrders: 3,
  createdAt: Date,
  updatedAt: Date
}
```

5. products

```
{
  _id: ObjectId,
  storeId: ObjectId,
  shopifyProductId: "gid://shopify/Product/123",
  handle: "custom-t-shirt",
  title: "Premium Custom T-Shirt",
  description: "...",
  vendor: "Demo Brand",
  productType: "T-Shirt",
  status: "active",
  options: [
    { name: "Color", position: 1, values: ["Black", "White", "Blue"] },
    { name: "Size", position: 2, values: ["S", "M", "L", "XL"] }
  ],
  images: [
    { shopifyImageId: "gid://shopify/ProductImage/100", url: "https://...", alt: "Black T-Shirt" }
  ],
  tags: ["customizable", "t-shirt"],
  customizerEnabled: true,
  shopifyCreatedAt: Date,
  shopifyUpdatedAt: Date,
  lastSyncedAt: Date,
  createdAt: Date,
  updatedAt: Date
}
```

6. product_variants

```
{
  _id: ObjectId,
  storeId: ObjectId,
  productId: ObjectId,
  shopifyVariantId: "gid://shopify/ProductVariant/456",
  title: "Black / XL",
  sku: "TS-BLK-XL",
  barcode: null,
  selectedOptions: { Color: "Black", Size: "XL" },
  price: 29.99,
  compareAtPrice: 34.99,
  currency: "USD",
  inventoryQuantity: 42,
  availableForSale: true,
  imageUrl: "https://...",
  status: "active",
  lastSyncedAt: Date,
  createdAt: Date,
  updatedAt: Date
}
```

7. customizer_configs

```
{
  _id: ObjectId,
  storeId: ObjectId,
  productId: ObjectId,
  name: "T-Shirt Customizer",
  version: 3,
  enabled: true,
  canvas: { logicalWidth: 1000, logicalHeight: 1200, backgroundColor: "#ffffff" },
  views: [
    {
      key: "front", label: "Front", sortOrder: 1, mockupAssetId: ObjectId,
      printArea: { x: 0.25, y: 0.18, width: 0.50, height: 0.60 },
      safeArea: { x: 0.28, y: 0.21, width: 0.44, height: 0.54 },
      bleed: { top: 0.02, right: 0.02, bottom: 0.02, left: 0.02 },
      physicalSize: { width: 12, height: 16, unit: "inch", dpi: 300 },
      enabled: true
    },
    {
      key: "back", label: "Back", sortOrder: 2, mockupAssetId: ObjectId,
      printArea: { x: 0.25, y: 0.18, width: 0.50, height: 0.60 },
      physicalSize: { width: 12, height: 16, unit: "inch", dpi: 300 },
      enabled: true
    }
  ],
  allowedTools: {
    uploadImage: true, text: true, shapes: true, clipart: true,
    qrCode: true, curvedText: false, backgroundRemoval: true, imageFilters: true
  },
  uploadRules: {
    allowedMimeTypes: ["image/png", "image/jpeg", "image/webp", "image/svg+xml"],
    maxFileSizeMb: 25, minimumWidth: 800, minimumHeight: 800, minimumDpi: 150
  },
  fontResourceIds: [ObjectId],
  clipartResourceIds: [ObjectId],
  pricingRuleSetId: ObjectId,
  defaultTemplateId: ObjectId,
  exportSettings: { format: "png", dpi: 300, transparentBackground: true, generateSvg:
true, generatePdf: false },
  createdBy: ObjectId,
  updatedBy: ObjectId,
  createdAt: Date,
  updatedAt: Date
}
```

8. design_resources

```
{
  _id: ObjectId,
  storeId: ObjectId,
  type: "clipart",
  name: "Classic Star",
  category: "Shapes",
  assetId: ObjectId,
  tags: ["star", "shape", "classic"],
  metadata: { fontFamily: null, fontWeight: null, previewUrl: "https://..." },
  visibility: "store",
  status: "active",
  createdBy: ObjectId,
  createdAt: Date,
  updatedAt: Date
}
```

9. pricing_rule_sets

```
{
  _id: ObjectId,
  storeId: ObjectId,
  name: "Standard T-Shirt Pricing",
  currency: "USD",
  rules: [
    {
      id: "front_print", name: "Front Print", type: "view_used",
      condition: { viewKey: "front", minimumObjectCount: 1 },
      calculation: { method: "fixed", amount: 0 }
    },
    {
      id: "back_print", name: "Back Print", type: "view_used",
      condition: { viewKey: "back", minimumObjectCount: 1 },
      calculation: { method: "fixed", amount: 5 }
    },
    {
      id: "background_removal", name: "Background Removal", type: "service",
      condition: { serviceKey: "background_removal" },
      calculation: { method: "fixed", amount: 3 }
    },
    {
      id: "quantity_discount", name: "Bulk Discount", type: "quantity",
      tiers: [
        { min: 10, max: 24, discountPercent: 5 },
        { min: 25, max: 49, discountPercent: 10 },
        { min: 50, max: null, discountPercent: 15 }
      ]
    }
  ],
  status: "active",
  createdAt: Date,
  updatedAt: Date
}
```

10. assets

```
{
  _id: ObjectId,
  storeId: ObjectId,
  customerId: ObjectId,
  designId: ObjectId,
  ownerType: "design",
  ownerId: ObjectId,
  type: "customer_upload",
  originalName: "logo.png",
  mimeType: "image/png",
  extension: "png",
  storageProvider: "s3",
  bucket: "product-customizer",
  storageKey: "stores/store-id/designs/design-id/logo.png",
  publicUrl: null,
  metadata: { width: 2500, height: 2500, dpi: 300, fileSize: 2450000, hasTransparency:
true, colorSpace: "sRGB", checksum: "sha256-hash" },
  processing: { status: "completed", virusScanStatus: "clean", backgroundRemoved: false,
optimizedAssetId: null },
  status: "active",
  expiresAt: null,
  createdAt: Date,
  updatedAt: Date
}
```

11. designs

```
{
  _id: ObjectId,
  storeId: ObjectId,
  customerId: ObjectId,
  guestSessionId: null,
  productId: ObjectId,
  variantId: ObjectId,
  customizerConfigId: ObjectId,
  title: "My Black T-Shirt Design",
  activeViewKey: "front",
  selectedOptions: { Color: "Black", Size: "XL" },
  status: "draft",
  currentRevision: 14,
  latestVersionId: ObjectId,
  viewSummary: [
    { viewKey: "front", objectCount: 4, previewAssetId: ObjectId, updatedAt: Date },
    { viewKey: "back", objectCount: 2, previewAssetId: ObjectId, updatedAt: Date }
  ],
  pricing: { productPrice: 29.99, customizationPrice: 5, discount: 0, total: 34.99,
currency: "USD" },
  services: [
    { key: "background_removal", assetId: ObjectId, price: 3, status: "completed" }
  ],
  lastAutosavedAt: Date,
  lastOpenedAt: Date,
  expiresAt: Date,
  createdAt: Date,
  updatedAt: Date
}
```

12. design_views

```
{
  _id: ObjectId,
  storeId: ObjectId,
  designId: ObjectId,
  viewKey: "front",
  revision: 14,
  canvas: {
    version: "6.0.0",
    width: 1000,
    height: 1200,
    objects: [
      {
        id: "object-uuid", type: "textbox", text: "Stay Positive",
        left: 240, top: 320, width: 520, height: 200,
        scaleX: 1, scaleY: 1, angle: 0,
        fill: "#000000", fontFamily: "Montserrat", fontSize: 72,
        metadata: { locked: false, printable: true, resourceId: null }
      }
    ]
  },
  canvasStorageAssetId: null,
  objectCount: 1,
  previewAssetId: ObjectId,
  validation: { hasLowResolutionAsset: false, hasOutOfBoundsObject: false, warnings: [] },
  updatedAt: Date,
  createdAt: Date
}
```

13. design_versions

```
{
  _id: ObjectId,
  storeId: ObjectId,
  designId: ObjectId,
  versionNumber: 3,
  source: "add_to_cart",
  label: "Cart Version",
  productSnapshot: { productId: ObjectId, shopifyProductId: "gid://shopify/Product/123",
title: "Premium Custom T-Shirt" },
  variantSnapshot: {
    variantId: ObjectId, shopifyVariantId: "gid://shopify/ProductVariant/456",
    title: "Black / XL", sku: "TS-BLK-XL", options: { Color: "Black", Size: "XL" }
  },
  pricingSnapshot: {
    basePrice: 29.99,
    charges: [{ key: "back_print", label: "Back Printing", quantity: 1, unitPrice: 5,
total: 5 }],
    discount: 0, total: 34.99, currency: "USD"
  },
  previewAssetIds: [ObjectId],
  locked: true,
  lockedAt: Date,
  createdBy: ObjectId,
  createdAt: Date
}
```

14. design_version_views

```
{
  _id: ObjectId,
  storeId: ObjectId,
  designVersionId: ObjectId,
  viewKey: "front",
  canvasJSON: { version: "6.0.0", objects: [] },
  canvasStorageAssetId: null,
  previewAssetId: ObjectId,
  printFileAssetId: null,
  physicalSize: { width: 12, height: 16, unit: "inch", dpi: 300 },
  checksum: "sha256-hash",
  createdAt: Date
}
```

15. cart_sessions

```
{
  _id: ObjectId,
  storeId: ObjectId,
  customerId: ObjectId,
  shopifyCartId: "gid://shopify/Cart/abc123",
  status: "active",
  lines: [
    {
      shopifyCartLineId: "gid://shopify/CartLine/line123",
      productId: ObjectId,
      variantId: ObjectId,
      designId: ObjectId,
      designVersionId: ObjectId,
      quantity: 1,
      pricingSnapshot: { basePrice: 29.99, customizationPrice: 5, total: 34.99 }
    }
  ],
  checkoutUrl: "https://...",
  expiresAt: Date,
  createdAt: Date,
  updatedAt: Date
}
```

16. orders

```
{
  _id: ObjectId,
  storeId: ObjectId,
  customerId: ObjectId,
  shopifyOrderId: "gid://shopify/Order/987",
  shopifyOrderNumber: "#1054",
  email: "customer@example.com",
  financialStatus: "paid",
  fulfillmentStatus: "unfulfilled",
  currency: "USD",
  subtotalPrice: 34.99,
  totalDiscount: 0,
  totalTax: 2.80,
  totalPrice: 37.79,
  shippingAddress: {
    firstName: "David", lastName: "Smith", address1: "123 Main Street",
    city: "New York", province: "NY", postalCode: "10001", country: "US"
  },
  productionStatus: "pending",
  placedAt: Date,
  cancelledAt: null,
  createdAt: Date,
  updatedAt: Date
}
```

17. order_items

```
{
  _id: ObjectId,
  storeId: ObjectId,
  orderId: ObjectId,
  shopifyLineItemId: "gid://shopify/LineItem/100",
  productId: ObjectId,
  variantId: ObjectId,
  designId: ObjectId,
  designVersionId: ObjectId,
  title: "Premium Custom T-Shirt",
  variantTitle: "Black / XL",
  sku: "TS-BLK-XL",
  quantity: 1,
  priceSnapshot: {
    basePrice: 29.99,
    customizationCharges: [{ key: "back_print", label: "Back Printing", amount: 5 }],
    unitPrice: 34.99, lineTotal: 34.99, currency: "USD"
  },
  previewAssetIds: [ObjectId],
  printFileAssetIds: [ObjectId],
  production: {
    status: "pending",
    assignedTo: null,
    history: [{ status: "pending", changedBy: null, changedAt: Date }]
  },
  createdAt: Date,
  updatedAt: Date
}
```

18. export_jobs

```
{
  _id: ObjectId,
  storeId: ObjectId,
  designId: ObjectId,
  designVersionId: ObjectId,
  orderItemId: ObjectId,
  bullJobId: "export-12345",
  type: "production_export",
  status: "queued",
  progress: 0,
  input: { viewKeys: ["front", "back"], format: "png", dpi: 300, transparentBackground:
true },
  output: { previewAssetIds: [], printFileAssetIds: [], manifestAssetId: null },
  attempts: 0,
  maxAttempts: 3,
  error: null,
  startedAt: null,
  completedAt: null,
  createdAt: Date,
  updatedAt: Date
}
```

19. webhook_events

```
{
  _id: ObjectId,
  storeId: ObjectId,
  shopifyWebhookId: "webhook-event-id",
  topic: "orders/create",
  shopDomain: "demo-store.myshopify.com",
  payloadHash: "sha256-hash",
  payloadAssetId: null,
  status: "received",
  attempts: 0,
  error: null,
  receivedAt: Date,
  processedAt: null,
  createdAt: Date,
  updatedAt: Date
}
```

20. audit_logs

```
{
  _id: ObjectId,
  storeId: ObjectId,
  userId: ObjectId,
  action: "order.production_status_changed",
  entityType: "order_item",
  entityId: ObjectId,
  before: { status: "pending" },
  after: { status: "in_production" },
  metadata: { ipAddress: "127.0.0.1", userAgent: "Chrome", requestId: "request-uuid" },
  createdAt: Date
}
```

Database Relationship Diagram

```
erDiagram
    STORES ||--o{ STORE_MEMBERS : has
    USERS ||--o{ STORE_MEMBERS : belongs_to
    STORES ||--o{ CUSTOMERS : has
    USERS ||--o{ CUSTOMERS : linked_to
    STORES ||--o{ PRODUCTS : owns
    PRODUCTS ||--o{ PRODUCT_VARIANTS : has
    PRODUCTS ||--o{ CUSTOMIZER_CONFIGS : configured_by
    CUSTOMIZER_CONFIGS }o--|| PRICING_RULE_SETS : uses
    STORES ||--o{ DESIGN_RESOURCES : owns
    CUSTOMERS ||--o{ DESIGNS : creates
    PRODUCTS ||--o{ DESIGNS : customized_as
    PRODUCT_VARIANTS ||--o{ DESIGNS : selected_in
    DESIGNS ||--o{ DESIGN_VIEWS : has
    DESIGNS ||--o{ DESIGN_VERSIONS : versions
    DESIGN_VERSIONS ||--o{ DESIGN_VERSION_VIEWS : contains
    DESIGNS ||--o{ ASSETS : uses
    DESIGN_VERSIONS ||--o{ EXPORT_JOBS : exported_by
    CUSTOMERS ||--o{ CART_SESSIONS : owns
    DESIGN_VERSIONS ||--o{ CART_SESSIONS : attached_to
    CUSTOMERS ||--o{ ORDERS : places
    ORDERS ||--o{ ORDER_ITEMS : contains
    DESIGN_VERSIONS ||--o{ ORDER_ITEMS : fulfills
    ORDER_ITEMS ||--o{ EXPORT_JOBS : requires
    STORES ||--o{ WEBHOOK_EVENTS : receives
    STORES ||--o{ AUDIT_LOGS : records
```

Critical Design Flow

Design Autosave Flow

```
Canvas change
  ↓
Frontend debounce
  ↓
Update design_views
  ↓
Increment designs.currentRevision
  ↓
Generate low-resolution preview
  ↓
Update designs.viewSummary
```

এই operation transaction-এর মধ্যে করা উচিত।

Add to Cart Flow

```
Validate current design
  ↓
Calculate price on backend
  ↓
Create design_versions
  ↓
Copy current states to design_version_views
  ↓
Mark version locked
  ↓
Create/update cart_sessions
  ↓
Add Design ID and Version ID to Shopify cart
```

Cart-এ mutable designId ছাড়াও locked designVersionId রাখতে হবে।

Order Creation Flow

```
Shopify orders/create webhook
↓
Verify HMAC
↓
Check webhook_events unique ID
↓
Create orders
↓
Create order_items
↓
Attach locked designVersionId
↓
Create export_jobs
↓
Generate production files
```

Transaction Boundaries

- **Transaction 1: Save Design**

Update design_views, Update designs.currentRevision, Update designs.lastAutosavedAt

- **Transaction 2: Lock Design Version**

Create design_versions, Create design_version_views, Update designs.latestVersionId, Update designs.status

- **Transaction 3: Order Webhook**

Create webhook_events, Create orders, Create order_items, Create export_jobs

MongoDB বনাম Object Storage

- **MongoDB-তে থাকবে:** User information, Product metadata, Design metadata, Pricing rules, Small canvas JSON, Order information, Job status, File metadata
- **S3/R2-তে থাকবে:** Customer uploaded images, Mockup images, Fonts, Clipart SVG, Large canvas JSON, Preview images, High-resolution print files, PDF, Production manifest
- **Redis-এ থাকবে:** BullMQ jobs, Cache, Rate-limit counters, Temporary sessions, WebSocket presence, Short-lived locks

Recommended Status Enums

```
type DesignStatus = "draft" | "saved" | "cart_locked" | "ordered" | "archived" | "expired";
type ExportJobStatus = "queued" | "processing" | "completed" | "failed" | "cancelled";
type ProductionStatus = "pending" | "design_validation" | "file_preparation" |
"ready_for_production" | "in_production" | "quality_check" | "packed" | "shipped" |
"delivered" | "cancelled";
```

Suggested Implementation Order

1. stores
2. users + store_members
3. products + product_variants
4. customizer_configs
5. assets
6. designs + design_views
7. design_versions + design_version_views
8. pricing_rule_sets
9. cart_sessions
10. orders + order_items
11. export_jobs
12. webhook_events
13. audit_logs

সবচেয়ে গুরুত্বপূর্ণ design decision হলো: editable design এবং ordered design snapshot আলাদা রাখা। Customer পরে design edit করলেও order-এর designVersionId অপরিবর্তিত থাকবে, ফলে production team সবসময় order করার সময়কার exact design file পাবে।